



MTE544 - AUTONOMOUS MOBILE ROBOTICS

UNIVERSITY OF WATERLOO

DEPARTMENT OF MECHANICAL AND MECHATRONICS ENGINEERING

Lab 1: Sensor data processing for mobile robots

Authors:

Hunter Adams (ID: 20997503)

Joe Chen (ID: your id number)

Jacob Wielowieyski (ID: your id number)

Date: October 8, 2026

Contents

1	Summary of the problem	3
2	Questions	3

Note:

This document is intended to provide a sample structure for the reports in ME202 at the University of Waterloo.

1 Summary of the problem



Figure 1: A goose.

Reference a Figure 1 or and equation (1).

$$\rho_l \left(R \ddot{R} + \frac{3}{2} \dot{R}^2 \right) = p_{gas} - P(t) - P_0 + \frac{R}{c} \frac{dp_{gas}}{dt} - 4\mu \frac{\dot{R}}{R} - \frac{2\sigma}{R} \quad (1)$$

For more contextual information on the bubble dynamic phenomena, please see the following sources: [?], [?], and [?].

2 Questions

This section answers the individual questions of the project description. For each question, provide an answer and short analysis.

Question 1

(a)

References to equations can be written out in latex (1). Similarly, figures 1 may also be referenced.

(b)

Citations require the mybib.tex file to be extended with the desired references. Students can use JabRef (<http://www.jabref.org>) to construct the mybib.bib file. Students are invited to link their Mendelay, CiteULike and Zotero account directly to Overleaf.

Students' contributions

Mr. Goose and Mrs. Goose worked together to understand the problem and write the numerical codes. The summary of the problem and Q2 were written up by Mr. Goose, Q1 and Q3 were completed by Mrs. Goose. Both students corrected the final report.